ENGINEERING DESIGN NOTE

Optical Physics & Geometric Rationale for Vertical Row Slice Extraction in PySpectrometer2

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1. Objective

This design note establishes the mathematical, optical, and sensor-level framework for modifying the open-source `PySpectrometer2` architecture to perform a 1D vertical row slice extraction. This alteration transforms the software from a horizontal dispersion tracker to a vertical dispersion tracker, optimizing the analysis of spatial light profiles when the optical train forces diffraction along the sensor's vertical axis.

2. The Rigid Body Reference Frame Problem

When restructuring a DIY spectrometer built around a rigid chassis (such as a 12-inch PVC tube housing an entry slit, a transmission/reflection diffraction grating, and a camera module), intermediate system updates can prompt non-intuitive spatial reference transformations. A common misconception is that a complete rotation of the external chassis by 90° relative to the lab bench will shift the dispersion orientation from horizontal to vertical inside the data processing pipeline.

Because the internal components are rigidly bound to one another, rotating the physical chassis spins the camera sensor in perfect tandem with the slit and grating. In the reference frame of the local pixel matrix, the dispersion vector remains aligned with the native horizontal X-axis. To map the physical spectrum across the sensor's vertical columns without programmatic modifications, an engineer would have to mechanically isolate the camera substrate and rotate it exactly 90° relative to the optical slit and grating alignment. Such a mechanical modification introduces stability risks and assembly complexities when using hand tools.

3. Optical Slit-Grating Alignment and Perpendicular Smearing

An alternative suggestion involves turning only the chassis end cap containing the entry slit by 90°. This violates a fundamental condition of transmission diffraction physics: **the long axis of the optical slit must remain precisely parallel to the micro-grooves of the diffraction grating.**

A diffraction grating scatters incident wave-fronts perpendicular to its internal ruling lines. When the entry slit is parallel to these lines, the light forms an array of clean, discrete spectral lines matching the geometry of the slit. If the slit cap is turned 90° while the grating remains stationary, the slit becomes perpendicular to the lines. Instead of constructive interference forming narrow line peaks, the spatial profiles overlap and smear across the sensor surface. The resulting cross-contamination destroys the signal-to-noise ratio (SNR), rendering noise-compensation techniques—such as wavelet transformations or simulated annealing—mathematically ineffective due to a lack of baseline peak definitions.

4. Sensor Architecture and Resolution Penalties

If the entry slit and the diffraction grating are rotated together as a unified sub-assembly while keeping the camera module stationary, the spectrum successfully projects along the vertical column array (***Y***-axis). However, executing this change via hardware rather than software incurs a severe resolution penalty due to the rectangular aspect ratio of modern solid-state sensors (e.g., CMOS/CCD matrices).

Sensor Orientation Dimension	Typical Pixel Count (e.g., Pi NoIR Module)	Spectral Resolution Impact (Wavelength Bins)	Calibration Stability Margin
Horizontal Axis (X-Width)	800 Bins	Maximum Sampling Density (~1.1 bin/nm)	High; accommodates slight structural pitch instabilities.
Vertical Axis (Y-Height)	480 Bins	Reduced Sampling Density (~0.6 bin/nm)	Low; slight yaw shifts cause the beam to miss the active pixel rows.

By shifting the dispersion axis to the shorter vertical dimension, the entire usable nanometer scale is compressed into a significantly smaller number of pixel bins. This reduction in discrete data points reduces the efficacy of digital processing routines that rely on high-density data arrays to resolve overlapping spectral lines.

5. Mathematical Signal-to-Noise Ratio (SNR) Optimization via Software Cross-Averaging

The standard horizontal implementation of `PySpectrometer2` reads a single vertical coordinate and samples across the width. If a hardware modification simply throws away adjacent spatial data, it introduces massive photon shot noise. By implementing a software column-averaging aperture window over the vertical slice, we unlock a powerful noise-suppression mechanism rooted in statistical mechanics.

Photon collection is governed by Poisson statistics, where the intrinsic shot noise of the signal scales as the square root of the number of detected photons (***N***). Thus, for a single un-averaged pixel profile, the baseline signal-to-noise ratio is given by:

$$SNR_{baseline} = rac{N}{\sqrt{N}} = \sqrt{N}$$

When the software isolates a vertical slice but integrates and averages across a small horizontal column aperture width of ***M*** pixels (e.g., `col_width = 5`), the total number of collected photons for that spectral bin increases to ***M imes N***. Assuming uncorrelated white noise and uniform shot noise across the narrow strip, the optimized signal-to-noise ratio transforms to:

$$SNR_{optimized} = rac{M \cdot N}{\sqrt{M \cdot N}} = \sqrt{M} \cdot \sqrt{N} = \sqrt{M} \cdot SNR_{baseline}$$

By averaging across a column width of $M = 5$ pixels, the signal-to-noise ratio is immediately multiplied by a factor of $\sqrt{5} \approx 2.236$, yielding a **123.6% net increase in signal clarity** before any advanced algorithmic processing occurs. This massive reduction in the noise floor provides the clean, high-fidelity data baseline required for downstream wavelet transformations and simulated annealing code to map target emission lines with sub-pixel precision.

6. Algorithmic Extraction Methodology

To preserve maximum spatial resolution, the hardware should remain anchored in its optimal horizontal aspect ratio. The shift from horizontal to vertical sampling is handled natively within the frame processing array. Instead of a standard horizontal reduction, the algorithm implements a vertical column slice with an integrated row-averaging window:

$$I(y) = \frac{1}{2w + 1} \sum_{x = x_c - w}^{x_c + w} F(x, y)$$

Where $I(y)$ represents the final 1D intensity profile vector across the vertical row index y, x_c is the targeted center column of the vertical slit image, w represents the aperture half-width for noise suppression, and $F(x, y)$ represents the raw pixel intensity value inside the monochrome frame buffer. Averaging across a narrow horizontal aperture strip ($col_width = 2w + 1 = M$) provides a running spatial average that suppresses high-frequency shot noise while preserving the full pixel row resolution of the vertical channel.